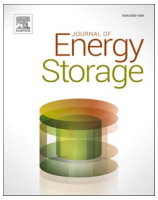
Journal of Energy Storage 73 (2023) 109247



Contents lists available at ScienceDirect

Journal of Energy Storage

journal homepage: www.elsevier.com/locate/est



Research papers



Numerical simulation and analysis of heat transfer and melting rate of nano-enhanced PCM composite embedded in a concentrator photovoltaic system

Atef Chibani [a,b,*], Slimane Merouani [b], Houssem Laidoudi [c], Aissa Dehane [b], Larbi Bendada [d], Leila Lamiri [a], Ghania Mecheri [e], Cherif Bougriou [f], Noureddine Gherraf [g]

[a] Research Center in Industrial Technologies CRTI, P.O.Box 64, Cheraga 16014, Algiers, Algeria
[b] Laboratory of Environmental Process Engineering, Department of Chemical Engineering, Faculty of Process Engineering, University Salah Boubnider Constantine 3, P.O. Box 72, 25000 Constantine, Algeria
[c] Faculty of Mechanical Engineering, USTO-MB, BP 1505, El-Menaouer, Oran 31000, Algeria
[d] Faculty of Sciences and Applied Sciences, Mechanical Engineering Department, Larbi Ben M'hidi University, 04000 Oum El Bouaghi, Algeria
[e] Faculty of Sciences and Applied Sciences, Department of Process Engineering, Larbi Ben M'hidi University, 04000 Oum El Bouaghi, Algeria
[f] Mechanical Engineering Department, Faculty of Technology, University of Mostefa Ben Boulaid, Batna 2, 05000 Batna, Algeria
[g] Laboratory of Natural Resources and Management of Sensitive Environments, Faculty of SE/SNV, Larbi Ben M'Hidi University, Oum El Bouaghi, Algeria

ARTICLE INFO

Keywords:
Nano-PCM
Melting evolution
Photovoltaic (PV)
Phase change materials (PCM)
Electrical efficiency

ABSTRACT

The primary objective of this research is to explore the impact of nanoparticles-infused PCM (phase change material) in the context of nano-PCM-PV technology (PV: photovoltaic panel). Computational investigations were conducted to evaluate the effectiveness of RT25HC, a paraffin wax PCM, in conjunction with various nanoparticles (MgO, TiO_2, ZnO and CuO) in solar panel cooling. The study also considers the influence of PV module inclination angles (β: 0 to 90°). High-resolution numerical simulations using the ANSYS-Fluent CFD platform were employed. The findings highlight the intricate relationship between nanoparticle addition, melting kinetics, electrical efficiency and PCM enthalpy. Specifically, the melting rate of RT25HC PCM exhibits notable differences between horizontal ($\beta = 0°$) and inclined panels ($\beta \neq 0°$), with a melting fraction of 12 % at $t = 2$ h for $\beta = 0°$ compared to 40–44 % for $\beta \neq 0°$. Nanoparticles maintain a higher electrical efficiency (11.6 %) for horizontal panels, whereas inclined panels ($\beta \neq 0°$) experience declining electrical efficiency (7.5 % at $t = 8$ h). For $\beta = 0°$, the PCM system effectively maintains the surface panel's temperature constant at 44 °C for a long time (up to 8 h). However, when nano-PCM systems are introduced, this temperature stabilizes at a slightly lower value of 40 °C, representing a 4 °C reduction attributed to the presence of nanoparticles. Conversely, for $\beta \neq 0$, the surface panel's temperature stabilizes at 44 °C for up to $t = 2$ h, then experiences a sharp increase, ultimately reaching 120 °C at $t = 8$ h, regardless of the type or absence of nanomaterials. Overall, the most effective application of nano-PCM for cooling the PV panel was predicted at a horizontal orientation of the panel, regardless of the nanomaterial type.

1. Introduction

The increased global demand for energy coupled with the environmental challenges posed by fossil fuels has propelled the worldwide focus on renewable energy sources. Consequently, governments across the globe have recognized the pivotal role of renewable energy, particularly solar energy, in their development plans [1–7]. While solar energy stands as a matured renewable energy source, it remains ripe for further exploration and advancement, particularly concerning the factors influencing PV efficiency and its sustainability impact [8–13]. The conversion efficiency and durability of PV solar cells are profoundly affected by excessive temperature rises. Traditional cooling systems, such as water and air, have been employed in photovoltaic-thermoelectric (PV-TE) coupling systems [14]. However, practical implementation faces restrictions due to the high costs and technical challenges [14].

Phase Change Materials (PCMs), encompassing a wide range of

* Corresponding author.
E-mail address: chibaniatef@gmail.com (A. Chibani).

https://doi.org/10.1016/j.est.2023.109247
Received 9 April 2023; Received in revised form 4 October 2023; Accepted 7 October 2023
Available online 17 October 2023
2352-152X/© 2023 Elsevier Ltd. All rights reserved.

Nomenclature

A_{mush}	mushy zone constant
Cp	heat capacity, (J/kg. K)
g	gravitational acceleration, (m/s^2)
K_B	Boltzmann constant
k	thermal conductivity, (W/ m K)
H	enthalpy of PCM,(J/kg. °C)
L	latent heat capacity, (kJ kg^{-1})
ΔH	(J/kg. K)
h	(J/kg. K)
h_p	(J/kg. K)
h_{ref}	(J/kg. K)
T	temperature, (°C)
L	latent heat of PCM, (J/kg)
q	Constant heat energy, (W/m^2)
n	number of temperature measurement point of PV cell along the x-direction
P	pressure, Pa
S	source term
u	velocity in the x-direction
v	velocity in the y-direction

Greek symbol

$\beta_{expansion}$	thermal expansion coefficient of liquid phase PCM, K^{-1}
ρ	density, kg/m^3
ɣ	liquid fraction of PCM
ε	small constant number
η	efficiency,%
η_{ref}	efficiency of PV cell under reference solar radiation %
β	tilt angle of the system, °
μ	dynamic viscosity, Pa.s
σ_t	temperature un evenness of the PV cell
Ø	volume concentration of nano-particles, %

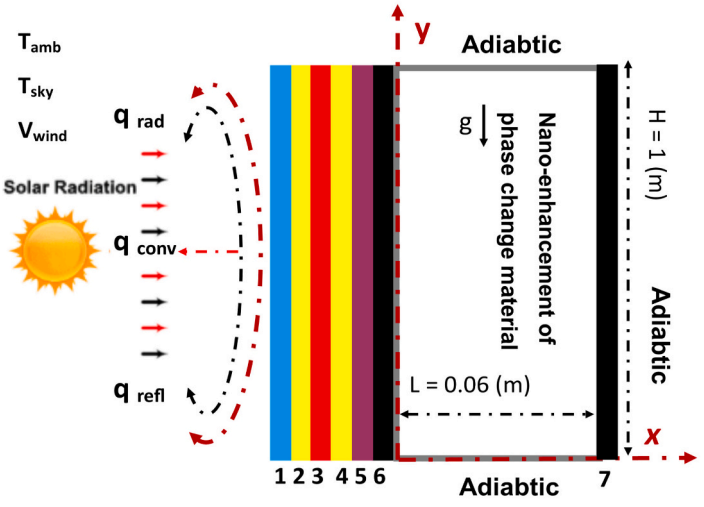
Subscript

ave	average
PCM	phase change material
Ref	reference
S	solid phase of PCM
l	liquid phase of PCM
e	electrical
th	thermal
x	x-direction of the system
y	y-direction of the system
np	nano-particles
npcm	nano-PCM
ESD	energy storage density
PTC	parabolic through collector
PV	photovoltaic
PVT	photovoltaic-thermal

substances such as paraffin wax, salt hydrates, organic compounds, and eutectic mixtures, play a pivotal role in enhancing the efficiency of solar panels [15]. PCMs possess the unique ability to manage thermal energy through latent processes, maintaining a fixed temperature [16]. This characteristic makes them an excellent choice for regulating the temperature of PV-TE systems, offering enhanced heat management [15,17–19]. Sharma [20] presented an up-to-date overview of PCM-integrated PV technology advances. Concentrating and non-concentrating PV modules integrated with PCMs are discussed in detail [20]. The heat recovered from photovoltaic panels is utilized in hybrid systems such as PCM integrated photovoltaic thermal (PVT/PCM) systems and thermoelectric modules integrated PCM regulated photovoltaic systems (PV/PCM-TE) for residential and industrial purposes [20]. The effects of ambient temperature, sunlight intensity, PCM melting point temperature, PCM channel orientation, various configurations to improve heat transfer, … etc., on the performance of PV modules were discussed [20].

On the other hand, many experimental and CFD calculations have attempted to provide more details about the characteristics of the PV/PCM systems with the improvement of PCM container performance in the presence of fins and metal foams. Xu et al. [21] experimentally examined the cooling characteristics of PCM on PV modules. They studied the effect of the reverse process of PCM from a molten state to a solidified state on the temperature of the cooled solar panel. They found that PCM can effectively reduce the temperature of solar modules under conditions of no wind, irradiance of 1000 W/m^2 and ambient temperature of 7.3 °C. PCM can reduce the average temperature of the top and back of the solar panel by 33.94 °C and 36.51 °C, respectively, within 300 min. In addition, PCM increased the average maximum power generation efficiency of solar photovoltaic modules by 1.63 %, and the average maximum output power increased by 1.35 W. On the other hand, solar PV modules without PCM cooling require only 60 min. For the cooling down of the Solar PV modules in a PV-PCM system to room temperature, 480 min was required. Lotfi et al. [22] considered two approaches to improve the efficiency of PV modules using phase change materials (PCMs) as coolants and integrated reflectors. The results show that the use of reflectors increases the radiation received by photovoltaic modules, but at the same time leads to an increase in the temperature of photovoltaic modules. A mirror tilt angle of 30° provides the best results. Jurcevic et al. [23] proposed and experimentally tested a photovoltaic thermal (PVT) collector in Mediterranean climate conditions. A comprehensive evaluation of PVT collector designs was carried out, considering performance, economic and environmental aspects. The specific PVT design consists of four cooling blocks, each equipped with straight pipes (water as the working medium) and a plexiglass container filled with an organic phase change material, i.e. pork fat. The examined systems achieved a maximum total energy efficiency of 62.2 %. However, the estimated annual average overall energy efficiency is expected to be below 50 %. Economic analysis shows that the average cost of the total energy generated by PVT collectors is between 0.056 and 0.083 V/kWh. Al-Nimr et al. [24] created models for phase change material storage, thermal regenerative electrochemical cycling, and concentrating photovoltaics (CPV/T). The concentrating photovoltaics immediately produce energy, heat water, and raise the temperature at which the thermal regeneration electrochemical circuit is charged throughout the day. Heat is stored in phase change materials in order to prevent heat loss to the environment and reduce the discharge temperature of the thermal regeneration electrochemical cycle. The heat from the storage tank is used once again throughout the night to raise the temperature at which the regeneration circuit is charged. The efficiency of the regenerative cycle and photovoltaic cells can reach 7.11 % and 15.74 % during the day. At night, the regenerative circuit is the only power source with a high efficiency of 6.13 %. Yusuf and Ballikaya [25] performed a numerical study of stand-alone high-concentration photovoltaic (CPV), CPV-PCM, CPV-TE, CPV-PCM-TE, and CPV-TE-PCM systems in the COMSOL Multiphysics environment. In addition to analyzing the system's energy efficiency, the study also examines the impact on the system's energy efficiency of using different PCMs and PCM containers with different internal configurations. The results showed that the energy conversion efficiencies of CPV-PCM, CPV-TE, CPV-PCM-TE, and CPV-TE-PCM were 34.8 %, 97.3 %, 106.5 %, and 111.4 % higher than monomeric CPV, respectively.

(a)



1. *Glass cover, (3 mm)*
2. *EVA, (0.5 mm)*
3. *Silicon cell, (0.5 mm)*
4. *EVA, (0.5 mm)*
5. *Tedlar, (0.3 mm)*
6. *Copper, (3 mm)*
7. *Steel, (5 mm)*

(b)

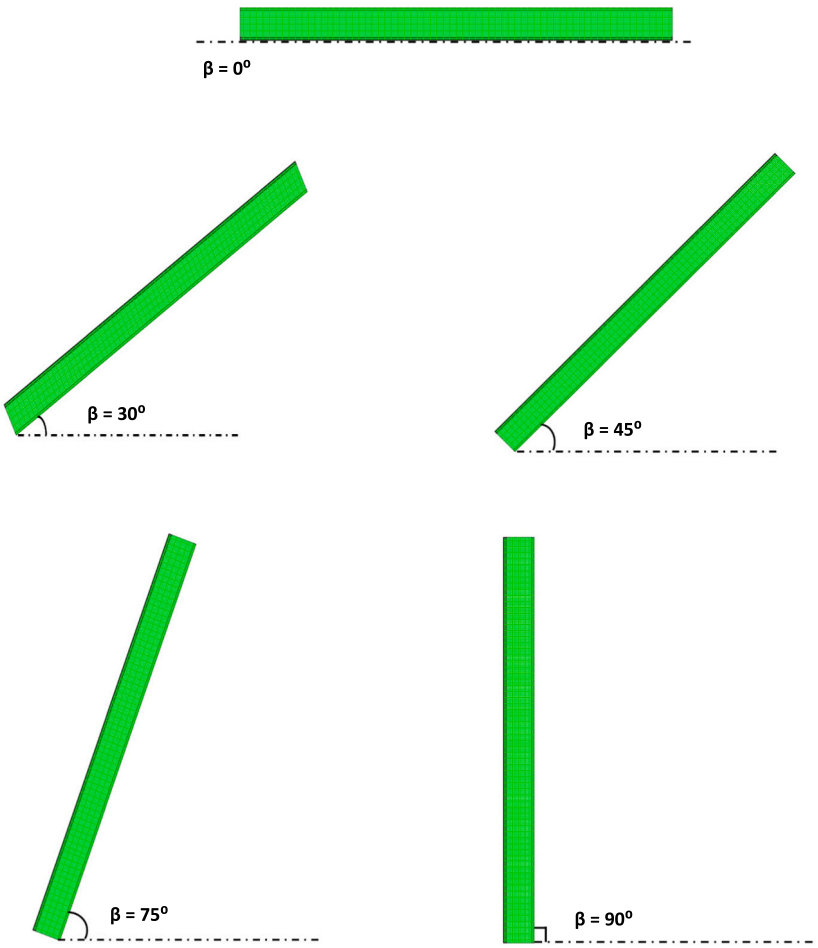


Fig. 1. Schematic of the physical model (a) and computational domain and grid structure for the Nano-PCM-Photovoltaic system (b).

Table 1
Thermophysical properties of chosen materials: PCM, metal foam and PV panel components [40,41,56–58].

Material	ρ (Kg/m^3)	λ (W/m K)	d_{np} nm	C_p (J/Kg K)	Latent heat (J/kg)	Phase change temperature (°C)
Magnesium Oxide (MgO)	3580	61.923	35	921	–	–
Titanium Oxide (TiO_2)	4250	08.954	100	686	–	–
Zinc Oxide (ZnO)	5630	27.196	100	494	–	–
Copper Oxide (CuO)	6500	17.991	15–30	536	–	–
PV	1190.667	0.2894	–	1552.006	–	–
Glass	3000	0.98	–	500	–	–
EVA	960	0.23	–	2090	–	–
Si	2330	148	–	677	–	–
Tedlar	1200	0.36	–	1250	–	–
Paraffin wax (RT25HC)	880/770	0.2	–	2000	240,000	22–26
Steel	7830	13	–	468	–	–
Copper	8933	350–401	–	381	–	–

Zhang et al. [26] conducted a numerical study on a solar photovoltaic PCM collector with rectangular copper fins and studied the effects of PCM height and fin length, arrangement, and number on the temperature characteristics and electrical efficiency of photovoltaic cells. The PCM height of 60 mm can meet the thermal management requirement of 800 W/m^2 heat flux input during the day. Compared with the finless PV-PCM model, the photovoltaic cell performance of the uniformly distributed three-fin model with a height of 33.33 mm was improved by 4.62 %. The eleven-fin model performed best, with average temperature, temperature non-uniformity, and efficiency of 48.58 °C, 3.20, and 11.19 %, respectively. These values are 16.11 °C lower, 43.06 % lower, and 7.70 % higher than the model without fins. Deng et al. [27] were looking for a way to characterize the distribution of pores in porous metal foams using fractal Brownian motion. A melting model for phase change materials was created and employed to examine the significance of pore distribution in the melting behavior of porous metal foams. The heat transmission process was remarkably improved by the use of porous metal foams.

Despite significant advancements in the field, it is noteworthy to indicate that the use of nano-PCM as a melting process enhancer (cooling enhancer) in PV-PCM strategies has received limited attention. Most researches have focused on the utilization of fins and metal foams. While the addition of nanoparticles to PCM has demonstrated considerable improvements in heat latent storage within thermal energy storage systems [28–31], its application within PV systems is absent.

Therefore, the primary objective of this study is to investigate the impact of incorporating nanoparticles into PCM within the context of nano-PCM/PV technology. To achieve this goal, we have conducted computational investigations to assess the effectiveness of RT25HC, a paraffin wax PCM, when combined with various nanoparticles (MgO, TiO_2, ZnO and CuO) in the cooling process of solar panels. This study also explores the influence of the inclination angle of PV modules, ranging from 0 to 90°. Our research methodology is based on high-resolution numerical simulations using the ANSYS-Fluent CFD platform.

2. Unit description, mathematical model and numerical method

2.1. Unit description

Fig. 1 shows a simplified scheme of the studied solar panel. The solar panel consists of 6 solid sheets placed on each other for the purpose of absorbing radiation and converting it into electricity, and a back container containing PCM. The purpose of the container with PCM is to absorb heat energy from the solid plates and thus reduce their temperature in order to increase the performance of the solar panel. The average solar irradiation was maintained at 800 W/m^2. The cross-section of the container has a length of 1 m and a width of 0.06 m (6 cm). The bottom side and both lateral walls of the container are adiabatic. The container is filled with solid PCM (or nano-PCM), which transitions into a liquid state when it absorbs a sufficient amount of heat. In the nano-PCM system, the particles are assumed to be perfectly mixed in paraffin, forming a single phase (nano-PCM phase), either in the solid or liquid phase. The physical properties of this phase are calculated using Eqs. (14)–(18), which differ from those of the pure PCM. In addition, this system is subjected to an inclination angle β from 0° to 90°, as shown in Fig. 1b. The thermophysical properties of the different system components (PV, sheets 1–6 materials, Paraffin RT25HC, nanoparticles) are regrouped in Table 1.

2.2. Mathematical model

Building upon the preceding description, our mathematical framework is articulated through the following equations [32]:

- Continuity equation

$$\frac{\partial \rho}{\partial t}+\frac{\partial(\rho u)}{\partial x}+\frac{\partial(\rho v)}{\partial y}=0 \tag{1}$$

In this equation, ρrepresents the density, u and v and denote the velocity in the and x, y directions, respectively.

- Momentum equations

$$\frac{\partial(\rho u)}{\partial t}+u\frac{\partial(\rho u)}{\partial x}+v\frac{\partial(\rho u)}{\partial y}=-\frac{1}{\rho}\frac{\partial P}{\partial x}+\mu\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}\right)+\rho g\beta_{expansion}cos\beta\left(T-T_{ref}\right)+S_x \tag{2}$$

$$\frac{\partial(\rho v)}{\partial t}+u\frac{\partial(\rho v)}{\partial x}+v\frac{\partial(\rho v)}{\partial y}=-\frac{1}{\rho}\frac{\partial P}{\partial y}+\mu\left(\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}\right)+\rho g\beta_{expansion}sin\beta\left(T-T_{ref}\right)+S_y \tag{3}$$

P is the pressure, μ is the dynamic viscosity, $\beta_{expansion}$ is the thermal expansion coefficient, and g represents gravitational acceleration of the PCM. β is the inclination of the collector. T_{ref} is the reference temperature, and S_x and S_y represent the source terms in the x- and y-directions, respectively [26].

The definitions of the source terms S_x and S_y are given as follows:

$$S_x=\frac{A_{mush}u(1-\gamma)^2}{(\gamma^3+\varepsilon)}S_y=\frac{A_{mush}v(1-\gamma)^2}{(\gamma^3+\varepsilon)} \tag{4}$$

In this study, the symbol γ represents the liquid fraction and A_{mush} is a constant value specific to the mushy zone, equal to 10^5 kg m^{-3} s^{-1}. Additionally, a small value ε is used to prevent division by zero, it is set to 0.001 [11,33,34].

- Energy equation

$$\frac{\partial(\rho H)}{\partial t}+u\frac{\partial(\rho H)}{\partial x}+v\frac{\partial(\rho H)}{\partial y}=k\left(\frac{\partial^2 T}{\partial x^2}+\frac{\partial^2 T}{\partial y^2}\right) \tag{5}$$

The enthalpy (H) of the PCM can be expressed as the sum of the

sensible enthalpy (h) and the latent heat enthalpy (ΔH) as follows [15,26,35]:

$$H = h + \Delta H \tag{6}$$

$$h = h_{ref} + \int_{T_{ref}}^{T} C_p dT \tag{7}$$

The enthalpy at the reference temperature T_{ref} is represented by h_{ref}. The latent heat enthalpy ΔH can be determined by using the latent heat L of the PCM through the following calculation [36,37]:

$$\Delta H = \gamma L \tag{8}$$

The liquid fraction **γ** is defined as follows [38]:

$$\gamma = \begin{cases} 0 & T < T_s \\ 1 & T > T_l \\ T - T_s/T_l - T_s & T_s < T < T_l \end{cases} \tag{9}$$

- Initial and boundary conditions

The starting temperature of the simulated area was established as 25 °C, at which point the PCM is fully solid. The initial conditions are defined as follows:

$$u = v = 0, T_{initial} = 25°C \tag{10}$$

The authors' previous experiment found that the thermal efficiency was around 80 % when the average solar radiation was 1000 W/m^2. Based on this, the 2D model used a constant heat flux boundary of 800 W/m^2. The boundary conditions for the model are summarized as follows:

The heat flow through the upper layer of the 2D model:

$$q = 800\ W/m^2 \tag{11}$$

The Other surrounding surfaces of the 2D model:

$$\left.\frac{\partial T}{\partial n}\right|_{x=0} = 0, \left.\frac{\partial T}{\partial n}\right|_{x=lp} = 0, \left.\frac{\partial T}{\partial t}\right|_{y=0} = 0 \tag{12}$$

The heat fluxes on both sides of the interface at the points where the metal fins, copper plate, and PCM are coupled are identical.

$$u = v = 0 \in$$
$$-k_{Cu}\frac{\partial T_{Cu}}{\partial n} = -k_{PCM}\frac{\partial T_{PCM}}{\partial n} \tag{13}$$

- Nano-PCM thermo-physical properties relations

To simulate the melting process of the PCM in the presence of metal-oxide nanoparticles, the following equations were used to obtain the thermos-physical properties (density, specific heat capacity, latent heat, and thermal expansion coefficient, respectively) of the nano-PCM system [39]:

$$\rho_{npcm} = (1-\varnothing)\rho_{pcm} + \varnothing\rho_{np} \tag{14}$$

$$cp_{npcm} = \frac{(1-\varnothing)\rho_{pcm}cp_{pcm} + \varnothing\rho_{np}cp_{np}}{\rho_{npcm}} \tag{15}$$

$$L_{npcm} = \frac{(1-\varnothing)\rho_{pcm}L_{pcm}}{\rho_{npcm}} \tag{16}$$

$$\beta_{npcm} = \frac{(1-\varnothing)\rho_{pcm}\beta_{pcm} + \varnothing\rho_{np}\beta_{np}}{\rho_{npcm}} \tag{17}$$

where **Ø** represents the nanoparticles volume fraction, $\mathbf{C_p}$ is specific heat

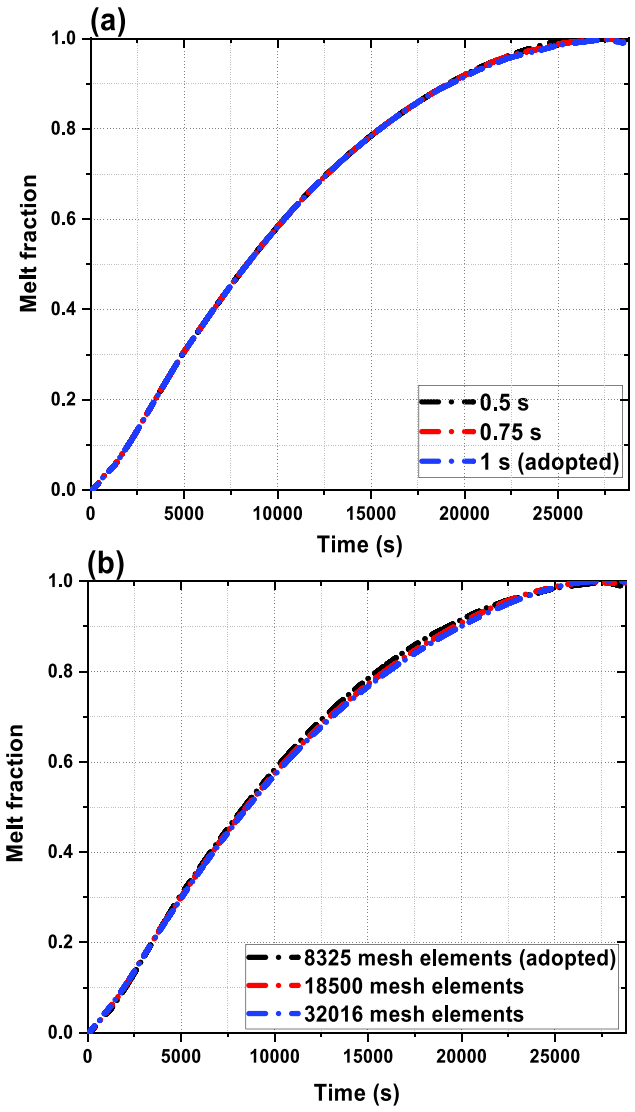


Fig. 2. Time step and grid size independency tests for of the average bed (CuO (5 %)-PCM system) temperature vs. time.

and the subscripts np, npcm, and pcm refer to the nanoparticle, nano-PCM, and pure PCM, respectively. The effective thermal conductivity of the nano-PCM, which includes the effects of particle size, particle volume fraction and temperature dependence as well as properties of the base PCM and the particle subject to Brownian motion, is given by [39–41]:

$$k_{npcm} = \frac{k_{np} + 2k_{pcm} - 2(k_{pcm} - k_{np})\varnothing}{k_{np} + 2k_{pcm} + (k_{pcm} - k_{np})\varnothing}k_{pcm} + 5.10^4\rho_{pcm}\varnothing cp_{pcm}\sqrt{\frac{K_B T_{pcm}}{\rho_{np} d_{np}}}f(T,\varnothing) \tag{18}$$

Where K_B is the Boltzmann constant, 1.381×10^{-23} J K^{-1}, and d_{np} is the particle diameter. The correction factor **f(T, ∅)** is defined as [39,42–44]:

$$f(T,\varnothing) = (2.8217 \times 10^{-2}\varnothing + 3.917 \times 10^{-3})\frac{T_{npcm}}{T_{ref}} + (-3.0669 \times 10^{-2}\varnothing - 3.91123 \times 10^{-3}) \tag{19}$$

T_{ref} (273 K) is the reference temperature. The effective dynamic viscosity is estimated as [40,41,45]:

$$\mu_{npcm} = 0.983e^{(12.958.\varnothing)}.\mu_{pcm} \tag{20}$$

(a)

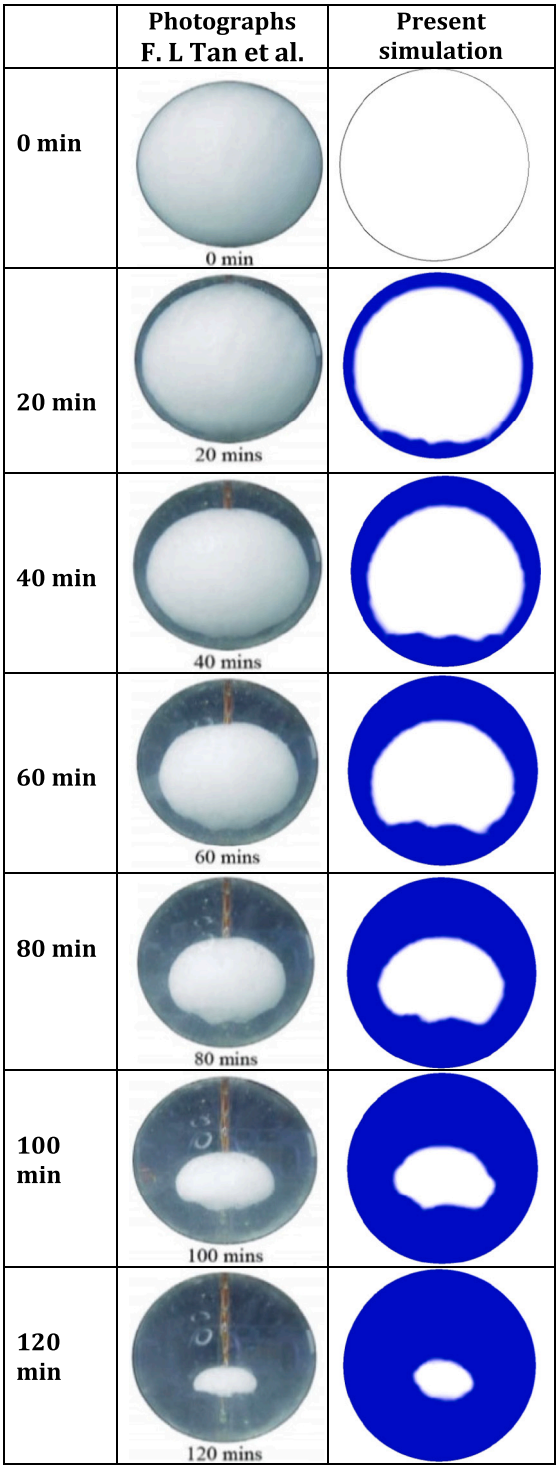


Fig. 3. Validation of the model with the numerical and experimental results of F. L Tan et al. [51] and M S. Mahdi et al. [50].

2.3. *Numerical procedure*

2.3.1. *Simulation parameters*

The Ansys fluent 15.0 commercial CFD software was used to find the numerical solution. The energy and momentum equations were solved using a second-order up-winding scheme, and the **PRESTO** Scheme was used for the pressure [26,46,47]. The **PISO** was employed to handle the coupling between velocity and pressure [40,48,49]. To improve stability during convergence, the under-relaxation factors for momentum, pressure, liquid fraction, and energy were set to 0.7, 0.3, 0.9 and 1 respectively [50]. The maximum number of iterations for each time step was set to 10, which met the convergence criteria for energy and velocity of 10^{-7} and 10^{-5} respectively.

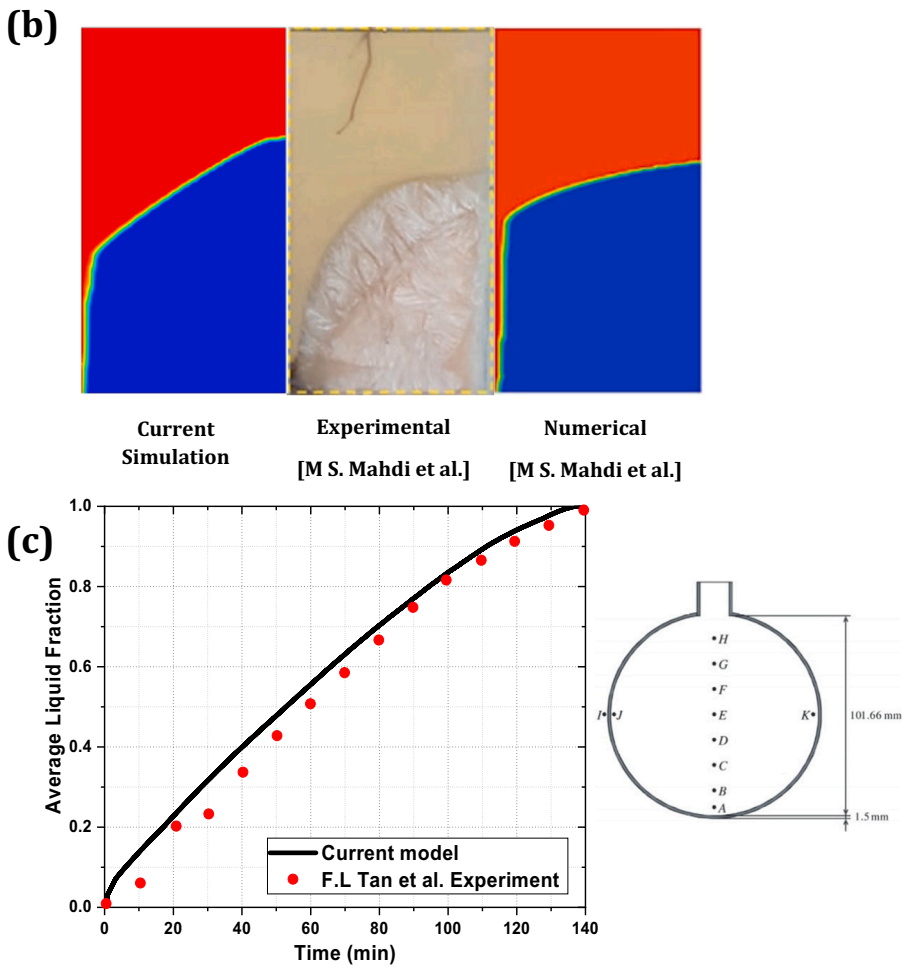


Fig. 3. (*continued*).

2.3.2. Time step and grid size independency tests

A study was conducted to determine the optimal number of cells for the mesh. Various cell numbers (8325, 18,500 and 32,016) were tested using a time step of 1 s, as shown in Fig. 2b. The results indicated that a mesh with 8325 cells is the most efficient for the current model, as there is minimal deviation when increasing the cell number to 32,016. This choice was made to save computational time. The same investigation was also applied to all relevant PV models, and a mesh of 8325 cells was selected due to their similar areas. Furthermore, the optimal time step was determined to be 1 s after additional testing with time step sizes of 0.5, 0.75 and 1 s on a mesh with 8325 cells (Fig. 1a).

2.3.3. Model validation

We diligently ensured the confidentiality of our mathematical model by rigorously validating it against many reported experimental data. This validation process involved comparing our simulation results with experimental data from Tan et al. [51] and both experimental and theoretical investigations by Mahdi et al. [50]. The side-by-side presentation of our findings alongside those of esteemed researchers is vividly depicted in Fig. 3a–c, serving as a critical benchmark for assessing the accuracy of our simulations.

Our model notably tracks the 2D melting progression of n-octadecane (PCM) over an extended melting period of up to 100 min, as vividly illustrated in Fig. 3a. While a slight deviation is discernible in comparison to the experimental and theoretical findings of Mahdi et al. [50] after 120 min of melting, as exhibited in Fig. 3b, our model demonstrates commendable precision overall. This minor discrepancy, particularly in the context of Lauric acid, underscores the inherent complexities of dynamic systems.

Expanding the scope of validation, we conducted a rigorous assessment of PCM melting fraction by comparing our model to an experimental curve from Tan et al. [51], as shown in Fig. 3c. Remarkably, our model aligns closely with the melting fraction evolution over 140 min. Both experimental and simulated liquid fraction profiles progressively evolve, reaching 100 % melting at 140 min for both, indicating a strong fit. The convergence of our simulation results and experimental data (Fig. 3c) underscores the robustness of our mathematical model in capturing the intricate dynamics of PCM melting.

3. Results and discussion

In this study, we investigate the cooling mechanisms in the PV panel shown in Fig. 1 using Paraffin wax (RT25HC) as the primary PCM. RT25HC was selected because its phase change transition occurs at approximately 26 °C, which aligns with the optimal working temperatures for PV panels [52]. To enhance the thermal properties of RT25HC, we introduced 5 % solid nanoparticles with the aim of improving heat transfer within the PV panel and optimizing its electrical efficiency. The selection of a 5 % nanoparticle concentration was based on previous studies [33,34], which demonstrated that this concentration level may offer optimal thermal enhancement of PCM properties without causing undesirable phenomena such as losing PCM weight as well as particle agglomeration and increased viscosity, which are considered constraints on heat transfer [53,54]. Our research involves a comprehensive examination of key parameters, including the nature of the added nanoparticles, the impact of time, and the angle of inclination of the PV panel.

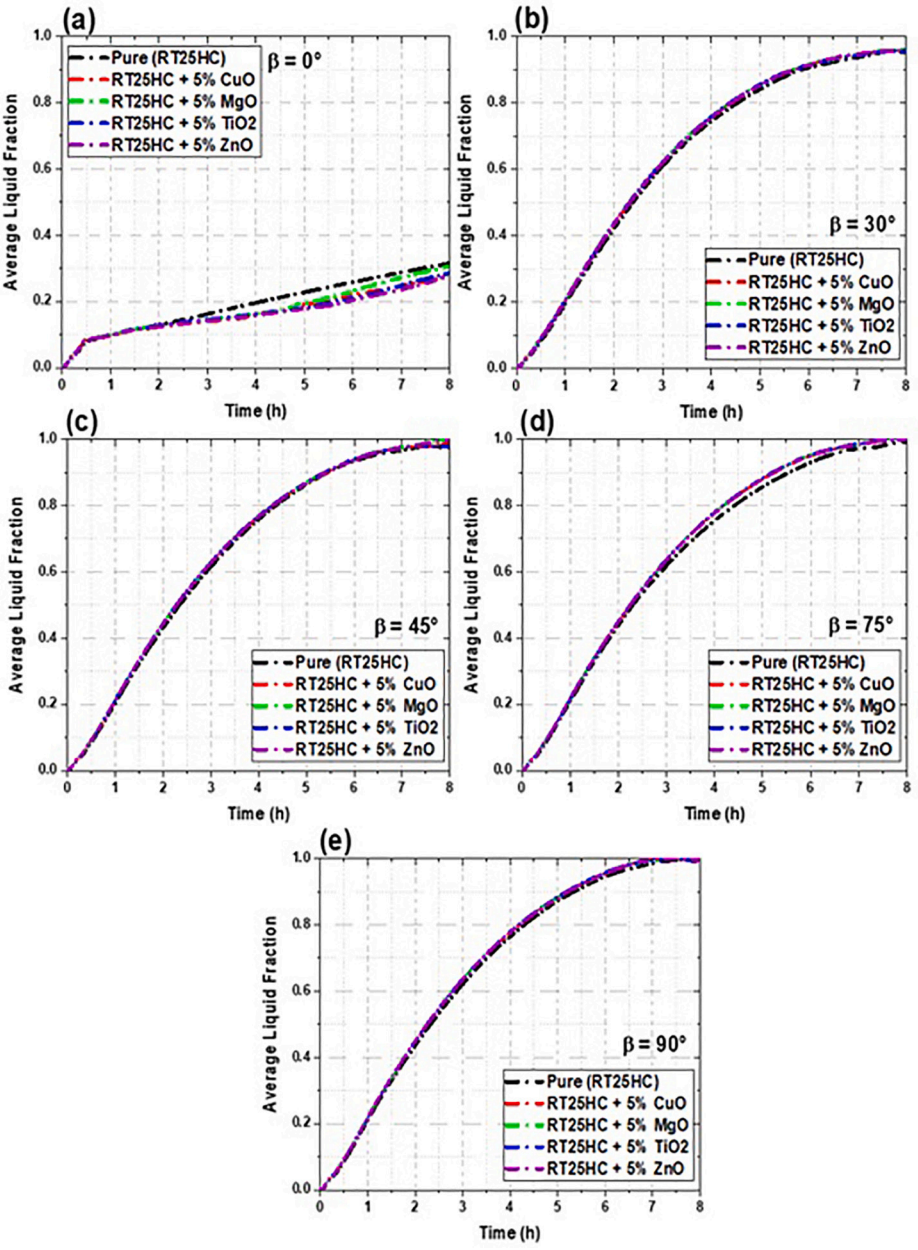


Fig. 4. Average PCM liquid fraction variation vs. time during the melting of RT25HC for different inclination angle (β) and nano oxides (5 %) type: CuO, MgO, TiO_2 and ZnO.

The selected nanoparticle types, specifically Magnesium Oxide (MgO), Titanium Oxide (TiO_2), Zinc Oxide (ZnO) and Copper Oxide (CuO), are outlined in Table 1. Additionally, we explored five different inclination angles (β) for the PV panels, ranging from 0° to 90°. This investigation provides insights into how these parameters collectively influence the PCM melting rate, electrical efficiency, system enthalpy and average temperature.

Before delving into the analysis of the results, it is important to consider some physical phenomena that help us to understand and discuss the new findings. Initially, thermal energy is transferred from the PV panel to the PCM (RT25HC) material through solid-to-solid contact (conduction mode). Once the temperature of the PCM reaches 26 °C (i.e. RT25HC phase change transition), the melting process begins. As the PCM transitions into a liquid state, another thermal phenomenon emerges, known as buoyancy-driven flow. In this mode, the hot liquid layers move counter to the force of gravity, while the cold layers move in the direction of gravity.

Also, in our study, the PCM does not typically reach a fully liquid state within the simulation time frame, as shown in Fig. 4. In cases where complete melting does occur, it takes more than 7 h to reach that state, and our simulations were stopped at 8 h (Fig. 4). As a result, the primary mode of heat transfer in our experiments predominantly involves conduction and natural convection within the PCM, as indicated by the results and simulations conducted. Therefore, while natural convection can occur to some extent, it does not dominate the heat transfer in our specific case due to the incomplete melting of the PCM. It's important to note that if the PCM achieves complete melting, it would transformed into a fully liquid state. In such a scenario, conduction heat transfer would still take place, but the dominant mode of heat transfer would likely shift towards natural convection due to the temperature gradient within the liquid PCM. This transition from a conduction-dominated heat transfer to natural convection is a complex process and would depend on various factors, including the shape of the container and the presence of nanoparticles.

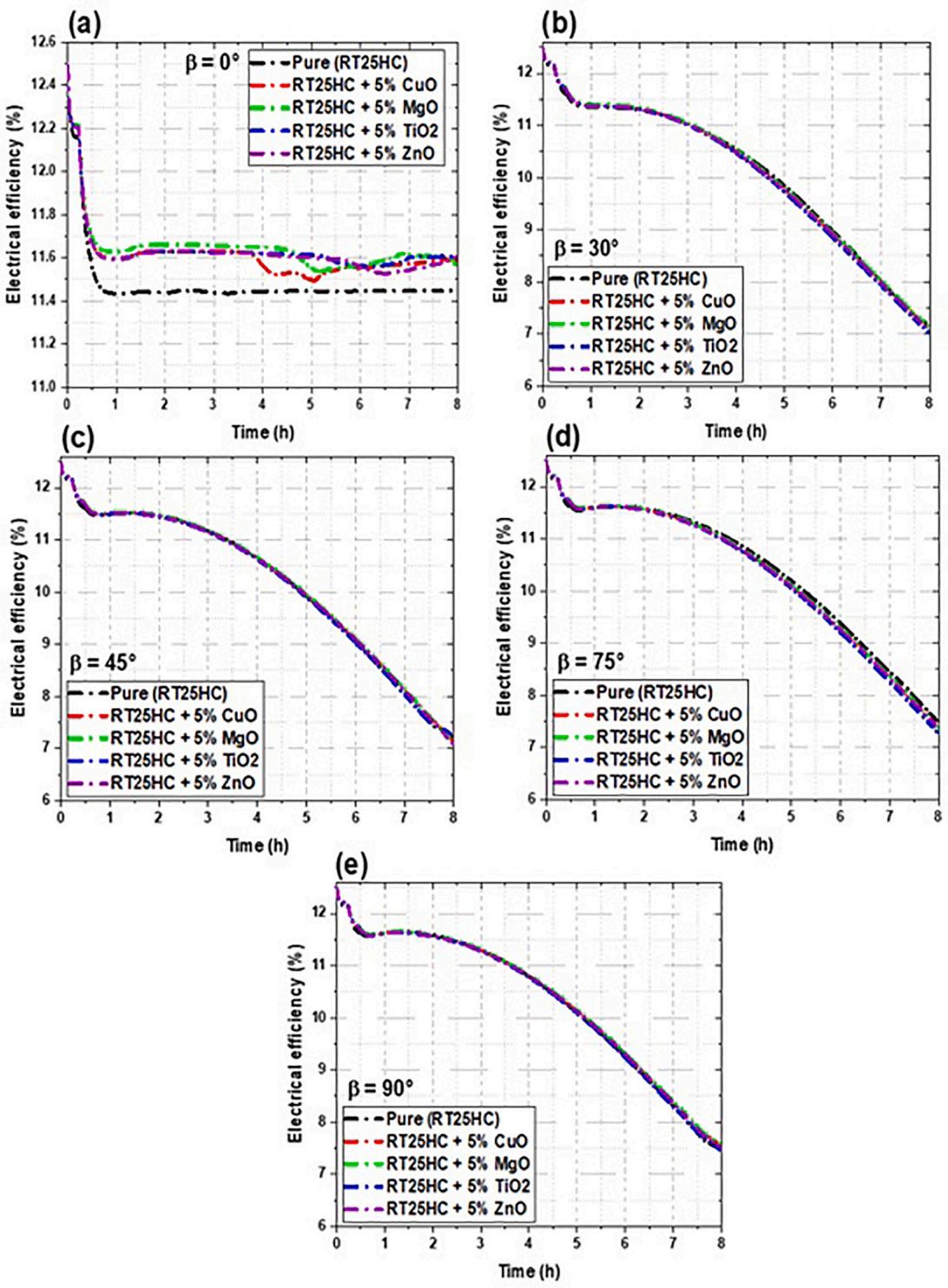


Fig. 5. PV-Electrical efficiency variation vs. time during the melting of RT25HC for different inclination angle (β) and nano oxides (5 %) type: CuO, MgO, TiO_2 and ZnO.

3.1. Effect of nano-oxides addition on the PCM melting fraction

In Fig. 4a-e, we observe the progression of melting kinetics measured as the average liquid fraction of RT25HC over a period of 8 h. We examine this progression both in the absence and presence of 5 % of various nanoparticles. These observations are conducted at different inclination angles of the panel: β = 0° (Fig. 4a), β = 30° (Fig. 4b), β = 45° (Fig. 4c), β = 75° (Fig. 4d), and β = 90° (Fig. 4e). It's important to note that as the inclination angle varies, the average liquid fraction of the PCM increases over time, but the rate of this increase differs based on the angle. The slowest melting rate is observed when the panel is horizontal (β = 0°) as depicted in Fig. 4a. This trend is in line with the obtained results by Abdulmunem et al. [55] on the effects of different tilt angles on the PCM melting in a rectangular container. In contrast, for inclination angles β = 30, 45, 75, and 90° (Fig. 4b-e), the melting fraction exhibits a consistent; independent evolution of β and nano-oxides

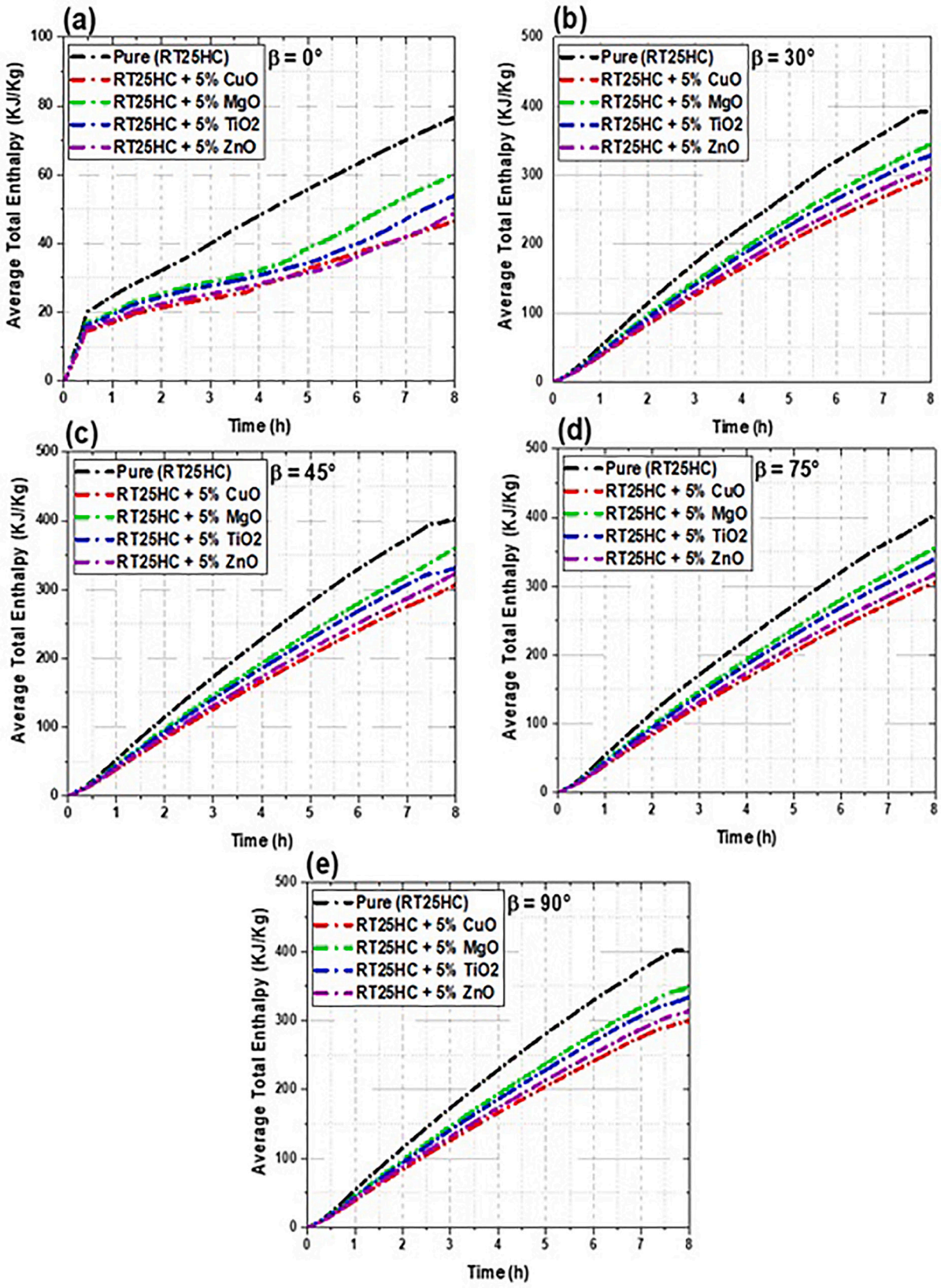


Fig. 6. Average total enthalpy variation vs. time during the melting of RT25HC for different inclination angle (β) and nano oxides (5 %) type: CuO, MgO, TiO_2 and ZnO.

presence and nature. Similar observations have been reported previously by another study [32], highlighting the influence of inclination angles on the melting process (without nanoparticles). For instance, at $t = 2$ h, the melting fraction reaches 12 % for a horizontal panel ($\beta = 0°$), while it ranges from 40 % to 44 % for all other angles ($\beta \neq 0°$). Over a longer duration at $t = 6$ h, the melting fraction increases to 20 % (25 % with nanoparticles) for the horizontal panel ($\beta = 0°$) and to 90–94 % for all other angles ($\beta \neq 0°$). Interestingly, the addition of nanoparticles at 5 % has a limited impact on the melting rate when the panel is in a horizontal position (about 5 % reduction). Conversely, there is virtually no noticeable effect of nanoparticle addition on the melting process for panels at angles other than horizontal. All these remarks are explained as follows:

(i) ***Lower Melting Rate in Horizontal Position***: The lower melting rate observed when the panel is in a horizontal position ($\beta = 0°$) can be attributed to the influence of gravity. In this orientation, the gravitational force acts directly against the movement of the hot liquid layers within the PCM. Consequently, it hinders the upward flow of hot liquid and the downward flow of cold liquid, which are essential for efficient heat transfer and phase change. This resistance to buoyancy-driven flow results in a slower melting rate.

(ii) ***Lack of effect on melting Rate for inclined panels ($\beta \neq 0°$)*:** In the case of inclined panels ($\beta \neq 0°$), the effect on the melting rate of the PCM is minimal. This is because, at these angles, gravity does not significantly impede the upward flow of hot liquid or the downward flow of cold liquid within the PCM. As a result, the gravitational force has little influence on the melting kinetics, leading to consistent and independent evolution of the melting fraction regardless of the inclination angle.

(iii) ***Limited impact of nanoparticle addition on the melting process*:** The addition of nanoparticles at a 5 % concentration does not significantly improve the melting rate of the PCM, especially in the case of inclined panels. This limited impact can be attributed to the fact that the primary heat transfer modes during PCM melting are conduction and natural convection, rather than convection driven by nanoparticles. In inclined panels, where natural convection is relatively unaffected by the presence of nanoparticles, their addition does not lead to a significant enhancement in heat transfer during the melting process.

In addition, the effect of nano-oxide addition can vary depending on specific factors such as operational conditions and the geometry and design of the panel. It may either enhance the melting process (due to increased PCM thermal conductivity) or inhibit it (due to increased viscosity of the PCM). The dominance of one of these mechanisms over the other depends on various factors, including reactor size and configuration. For example, in the case of a small rectangular PCM bed, the enhanced effect of nano-PCMs has been observed, while in large-scale reactors, their impact on heat exchange has been reported to be insignificant. In our case, involving an inclined PV panel, it appears that we strike an optimal balance between the enhancing and decelerating effects of nanoparticle addition, resulting in a negligible overall impact on the melting rate of the PCM.

(iv) ***Slight Reduction in Melting Rate with Nanoparticle Addition for Horizontal Panel*:** Interestingly, when nanoparticles are added at 5 % to a horizontal panel, there is a minor reduction in the melting rate (by about 5 % at $t = 8$ h, Fig. 4a). This can be attributed to the potential increase in viscosity caused by the presence of nanoparticles. An increase in viscosity can impede the flow of the PCM and slightly reduce the rate at which it melts when the panel is horizontal. However, this effect is relatively small and not as pronounced as the impact of gravity on melting kinetics in this orientation.

3.2. Effect of nano-oxides addition on panel electrical efficiency

The panel's electrical efficiency responses under various scenarios, as discussed in the context of melting fractions (Fig. 4a-e), are presented in Fig. 5a-e. For the horizontal panel ($\beta = 0°$), the electrical efficiency showed a decline from an initial value of 12.5 % to 11.43 % during the first 30 min. Subsequently, it stabilized at this lower value throughout the 8-h simulation period. In contrast, when nano-PCM systems were introduced, the electrical efficiency dropped from 12.5 % to only 11.6 %, and this value remained stable, regardless of the nano-material type. This difference in behavior observed for $\beta = 0°$ can be attributed to the role of nanoparticles in enhancing thermal conductivity within the PCM. In the case of the horizontal panel ($\beta = 0°$), where natural convection is limited due to the absence of an inclination angle, the addition of nanoparticles enhances heat transfer efficiency within the PCM material. This helps in maintaining a slightly higher electrical efficiency compared to the pure PCM.

For panels with an inclination angle ($\beta \neq 0°$), notably different behavior was observed. In these cases, the electrical efficiency dropped from 12.5 % to 11.6 % during the initial 30 min and stabilized at 11.6 % for the first 2 h. However, beyond this point, there was a near-linear decline in electrical efficiency, reaching 7.5 % at the end of the 8-h simulation. It should be noted that the observed results for $\beta \neq 0°$ are independent of the nanomaterial type. Consequently, the added nanoparticles have only a small effect on the electrical efficiency of the solar panel when the inclination angle of the panel is 0°.

The variation in electrical efficiency between horizontal and inclined panels can be attributed to the combined effects of natural convection and nanoparticle-enhanced heat transfer. In inclined panels, natural convection plays a more significant role due to the inclination angle, which promotes the movement of heated fluid layers. As a result, the beneficial effect of nanoparticles on heat transfer becomes less pronounced. This, coupled with the gravitational forces affecting natural convection, leads to a steeper decline in electrical efficiency over time for inclined panels compared to horizontal panels.

For $\beta = 0°$, it appears an inverse effect of the nano-systems on the melting rate (limited reduction, Fig. 4a) and the electrical efficiency (relatively higher value compared to pure PCM, Fig. 5a). This seemingly inverse effect was attributed to the distinct roles that nanoparticles play in heat transfer processes within the system. In the case of the melting rate, the primary heat transfer mechanisms involved are conduction and natural convection. Nanoparticles, while capable of enhancing thermal conductivity, have a relatively limited impact on these mechanisms. As a result, their contribution to speeding up the phase change process (melting rate) remains modest. However, when considering electrical efficiency, the role of nanoparticles becomes more pronounced. By improving thermal conductivity, nanoparticles aid in the efficient transfer of heat generated by the PV panel to the PCM. This, in turn, helps maintain lower temperatures within the PCM, which is critical for optimizing electrical efficiency. Thus, while nanoparticles may not significantly affect the rate of phase change, their positive influence on thermal conductivity benefits the overall electrical performance of the panel by keeping temperatures lower. This nuanced interplay between heat transfer mechanisms underscores the intricate relationship between nanoparticle addition and the thermal behavior of the PCM in solar panels.

3.3. Effect of nano-oxides addition on PCM enthalpy

The evolution of nano-PCM total enthalpy under various scenarios is depicted in Fig. 6a-e. Enthalpy is closely tied to temperature and the motion of particles, so it naturally increases with time as temperature rises. The influence of both the inclination angle (β) and the type of nanoparticles on total enthalpy is readily apparent. For $\beta = 0°$, the enthalpy of pure PCM experiences a rapid increase to 20 KJ/kg (15–20 KJ/kg for each nano-system) within the initial 30 min. Subsequently, it

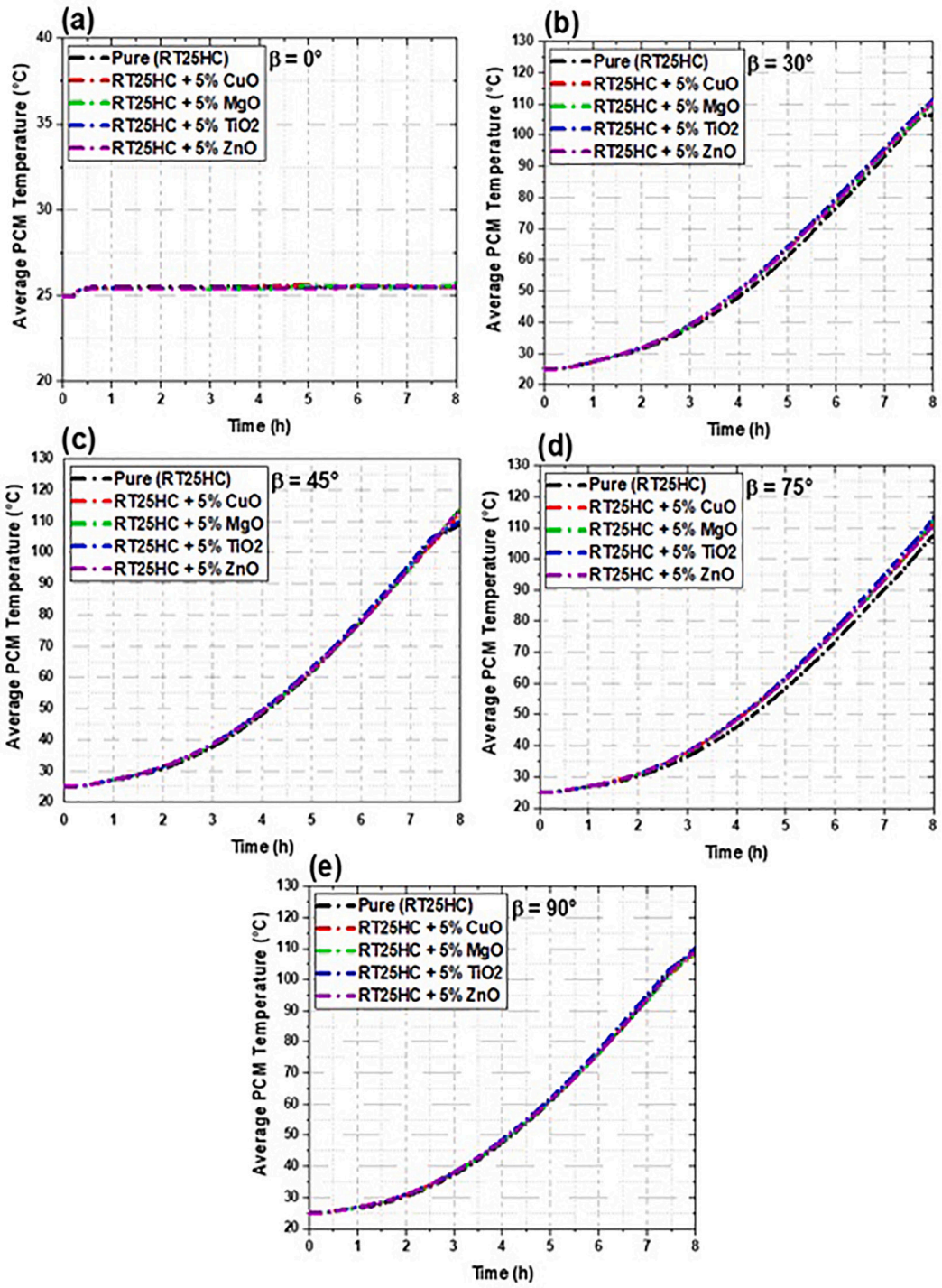


Fig. 7. Average PCM temperature variation vs. time during the melting of RT25HC for different inclination angle (β) and nano oxides (5 %) type: CuO, MgO, TiO_2 and ZnO.

continues to rise linearly, reaching 70 KJ/kg at $t = 8$ h. The impact of nanoparticle type becomes evident at this point: 60 KJ/kg for MgO, 55 KJ/kg for TiO_2, 50 KJ/kg for ZnO and 45 KJ/kg for CuO. On the other hand, the nanoparticle effect is consistent across various inclination angles (30°, 45°, 75° and 90°), all exhibiting a similar trend. In these cases, total enthalpy continues to increase, reaching 400 KJ/kg for pure paraffin and 350 KJ/kg, 330 KJ/kg, 310 KJ/kg and 300 KJ/kg for the MgO, TiO_2, ZnO and CuO-based nano-systems, respectively.

Thus, the presence of nanoparticles in the PCM leads to a reduction in stored enthalpy, primarily attributed to the increase in viscosity within the PCM material. This increase in viscosity acts as a hindrance to effective heat transfer across the PCM body. Among the nanoparticles studied, CuO had the most pronounced effect in reducing enthalpy, likely due to its lower thermal conductivity compared to MgO and ZnO. Interestingly, the trend in the reduction of enthalpy aligns with the thermal conductivity of the nanoparticles, except for TiO_2. For instance, MgO, with a higher thermal conductivity of 61.92 W/m K, had a relatively smaller impact on enthalpy reduction compared to CuO, which has a lower thermal conductivity of 17.99 W/m K. This suggests that the nanoparticles' influence on enthalpy is influenced by their thermal

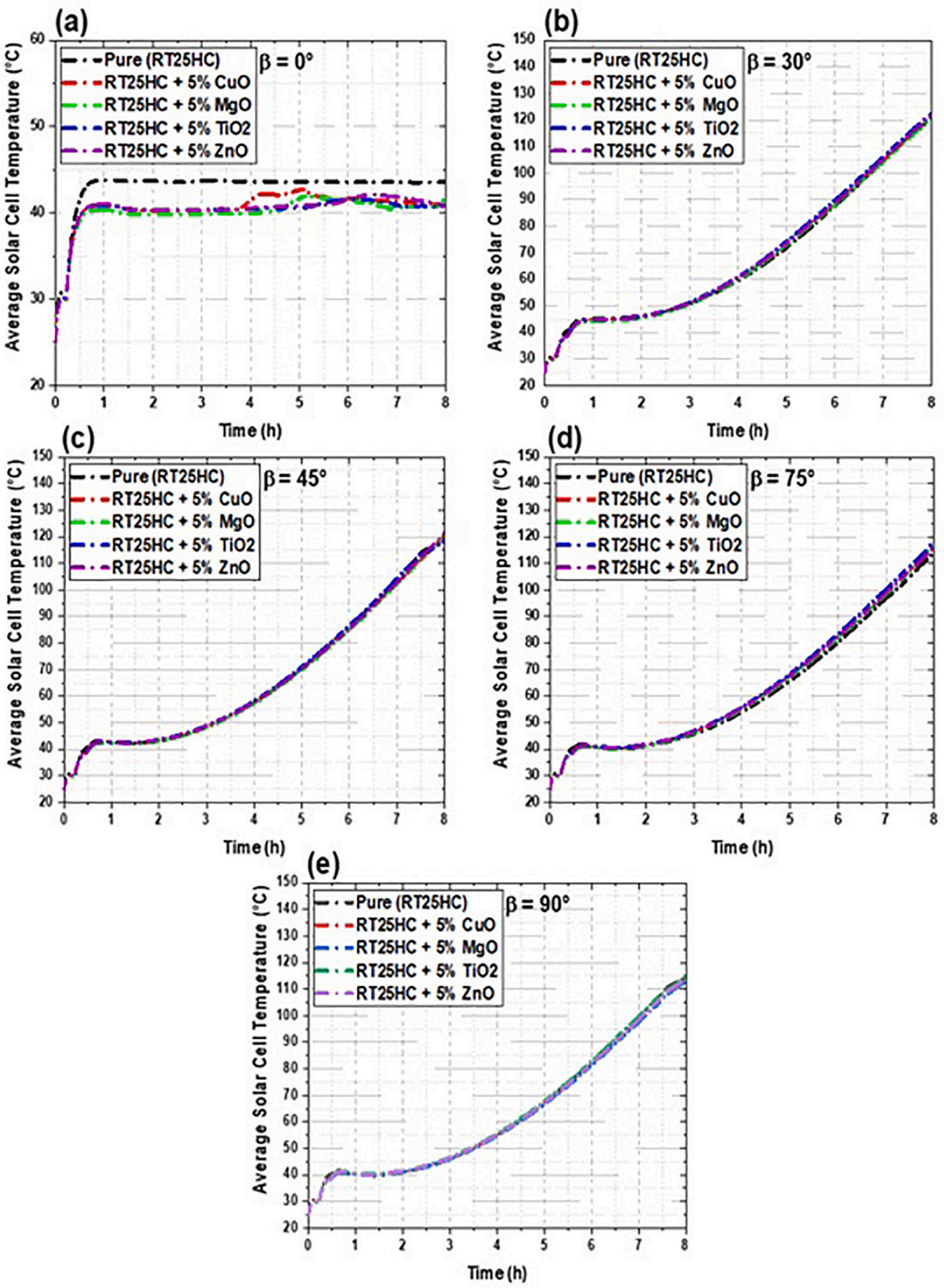


Fig. 8. Average solar cell temperature variation vs. time during the melting of RT25HC for different inclination angle (β) and nano oxides (5 %) type: CuO, MgO, TiO_2 and ZnO.

conductivity, except for the unique behavior exhibited by TiO_2.

3.4. Effect of nano-oxides on PCM and PV cell temperatures

The evolution of the average PCM temperature is illustrated in Fig. 7a-e, providing insights into its behavior under different conditions. Fig. 7a reveals that, for a horizontal panel ($\beta = 0°$), the mean PCM temperature remains relatively constant at around 26 °C. This stability arises from the continuous melting transition phase in this scenario, characterized by a higher percentage of the solid phase. In contrast, for inclined panels ($\beta \neq 0$), a notable increase in the mean temperature is observed over time. Specifically, the temperature reaches 30 °C at $t = 2$ h and 110 °C at $t = 8$ h, regardless of the type of nano-PCM system used.

These PCM temperature trends align with the findings from earlier analyses. The constant temperature observed for $\beta = 0°$ corresponds to a slower melting rate, which contributes to the efficient cooling of the solar panel, as demonstrated by the higher electrical efficiency observed in this scenario. Conversely, for inclined panels ($\beta \neq 0$), the continuous temperature rise reflects a faster melting process and a decline in electrical efficiency over time. Additionally, the variation in PCM

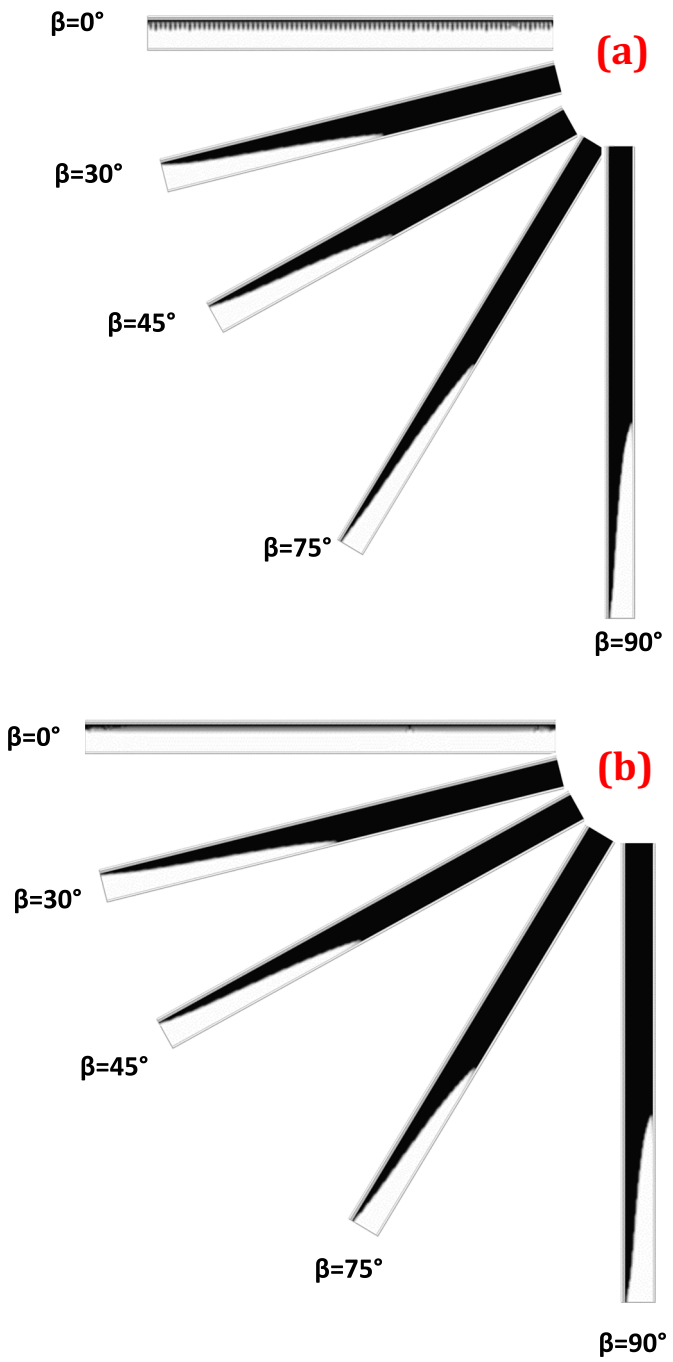


Fig. 9. Melt front of PCM at t = 12,000 s for different inclination angles. (a): pure RT25HC, (b) MgO(5 %)-RT25HC. Black color: liquid part of PCM, White color: solid part.

temperature correlates with the patterns observed in the liquid fraction contours, where a slower melting rate leads to a more favorable cooling effect. The findings underscore the intricate relationship between melting kinetics, electrical efficiency, and temperature evolution in the context of nano-PCM-enhanced solar panels.

Fig. 8a-e display the evolution of the average PV cell temperature, corresponding to the same conditions as those explored in Fig. 7a-e. In the specific case of horizontal panels (β = 0), as seen in Fig. 8a, the PCM system effectively maintains the PV cell temperature at a constant 44 °C for a long time (up to 8 h). However, when nano-PCM systems are introduced, this temperature stabilizes at a slightly lower value of 40 °C, representing a 4 °C reduction attributed to the presence of nanoparticles.

Conversely, for inclined panels (β ≠ 0), as shown in Fig. 8b-e, the temperature evolution follows a pattern akin to that observed in different PCM temperature scenarios (Fig. 7b-e). Initially, the PV cell temperature stabilizes at 44 °C for up to t = 2 h. Subsequently, it experiences a sharp increase, ultimately reaching temperatures as high as 120 °C at t = 8 h, regardless of the type or absence of nanomaterial. This intricate temperature behavior can be comprehended by considering the interplay between several factors, including melting kinetics, PCM temperature, electrical efficiency, and PV cell temperature. In scenarios where the PCM temperature remains stable and the melting process proceeds relatively slowly, such as in the case of β = 0 and nano-PCM-enhanced systems, the PV cell temperature remains lower, leading to higher electrical efficiency. Conversely, for inclined panels (β ≠ 0), the faster melting kinetics drive a significant rise in PV cell temperature over time, resulting in elevated temperatures irrespective of the utilization of nano-PCM. These observed temperature trends underline the complex relationship between melting kinetics, PCM properties, electrical efficiency, and PV cell temperature, all of which collectively impact the overall performance and thermal behavior of the solar panel system.

3.5. Melting contours

Fig. 9a-b illustrates the contours of liquid fraction concerning the inclination angle β at a time of 3.33h. Fig. 9a represents pure RT25HC (without any nanoparticles), while Fig. 9b represents MgO-RT25HC. In these plots, the black color indicates the liquid state of the PCM, whereas the white color represents the solid state. Upon comparing the contours of Fig. 9a and Fig. 9b, the impact of MgO nanoparticles on the liquid fraction appears to be relatively minor, with the exception of the case where β = 0, where a slight change in the liquid fraction rate is noticeable. Conversely, the inclination angle β exhibits a significant influence on the melting rate, where a larger angle β corresponds to a higher rate of liquid fraction, particularly when contrasting β = 0 with β ≠ 0. This behavior can be attributed to the movement of PCM liquid particles induced by thermal buoyancy forces, thereby enhancing heat transfer from the PV to the PCM. Importantly, the observations in Fig. 9a-b align with the trends observed in previous figures (especially Figs. 4-6), wherein a slower melting process corresponds to more effective cooling of the solar panel.

In conclusion, the contour findings presented in Fig. 9a-b corroborate the trends observed throughout the study. They reaffirm that the rate of PCM melting plays a crucial role in the cooling process of the solar panel. The slight influence of MgO nanoparticles on the liquid fraction, except at β = 0, underscores the nuanced relationship between nanoparticle addition and the thermal behavior of the PCM in solar panels.

4. Conclusion

In this study, we set out to explore the impact of incorporating nanoparticles into phase change materials (PCM) within the context of nano-PCM/PV (photovoltaic) technology. Through comprehensive computational investigations using ANSYS-Fluent CFD simulations, we have shed light on critical aspects of this novel approach.

Our findings have unveiled a nuanced interplay of factors. For instance, the addition of nanoparticles, including MgO, TiO_2, ZnO and CuO, demonstrated varying effects on the melting rates of PCM within different PV panel inclinations. Notably, in the case of horizontal panels (β = 0°), the PCM system effectively maintained the panel's surface temperature at 44 °C for extended durations, with a slight reduction to 40 °C observed when nano-PCM systems were introduced. Conversely, for inclined panels (β ≠ 0°), the temperature evolution followed a distinct pattern, stabilizing at 44 °C for up to 2 h, then sharply rising to 120 °C at t = 8 h, regardless of the presence or absence of nanomaterials. Furthermore, our study delved into the electrical efficiency of the panels, revealing that nanoparticles maintained higher electrical efficiency (11.6 %) for horizontal panels, while inclined panels (β ≠ 0°)

experienced declining efficiency, stabilizing at 7.5 % at t = 8 h, regardless of nanoparticle type.

In conclusion, our research enhances our understanding of nano-PCM applications in solar PV panels. The best application of nano-PCM for cooling the PV panel was observed in a horizontal panel orientation, irrespective of the nanomaterial type used. This insight is pivotal in advancing the efficiency and sustainability of solar panel technology.

As we continue to push the boundaries of solar panel technology, future research could explore additional parameters and materials to optimize the performance of nano-PCM systems. Investigating the effects of different nanomaterial compositions, such as hybrid nanomaterials, and their impacts on melting rates and electrical efficiency could offer valuable insights. Additionally, the integration of smart control systems to dynamically adjust the use of nano-PCM based on varying environmental conditions is a promising avenue. These advancements have the potential to contribute significantly to the evolution of solar energy technology and its role in sustainable energy solutions for the future.

CRediT authorship contribution statement

Atef Chibani: Conceptualization, Software, Methodology, Original draft preparation, Validation.

Slimane Merouani: Visualization; Data analysis, Writing-Reviewing and Editing, Supervision.

Houssem Laidoudi: Visualization; Data analysis, Writing-Reviewing and Editing.

Aissa Dehane: Visualization, Writing-Original draft, preparation, Validation.

Larbi Bendada: Data analysis, Writing-Reviewing and Editing, Methodology.

Leila Lamiri: Data analysis, Writing-Reviewing and Editing.

Ghania Mecheri: Data analysis, Writing-Reviewing and Editing.

Cherif Bougriou: Data analysis, Writing-Reviewing and Editing.

Noureddine Gherraf: Data analysis, Writing-Reviewing and Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

[1] J. Emhofer, K. Marx, A. Sporr, T. Barz, B. Nitsch, M. Wiesflecker, W. Pink, Experimental demonstration of an air-source heat pump application using an integrated phase change material storage as a desuperheater for domestic hot water generation, Appl. Energy 305 (2022) 117890, https://doi.org/10.1016/j.apenergy.2021.117890.

[2] H.M. Teamah, Comprehensive review of the application of phase change materials in residential heating applications, Alex. Eng. J. 60 (2021) 3829–3843, https://doi.org/10.1016/j.aej.2021.02.053.

[3] Y. Wang, J. Bailey, Y. Zhu, Y. Zhang, S.K.S. Boetcher, Y. Li, C. Wu, Application of carbon nanotube prepared from waste plastic to phase change materials: the potential for battery thermal management, Waste Manag. 154 (2022) 96–104, https://doi.org/10.1016/j.wasman.2022.10.003.

[4] S. Mandal, S. Ishak, D.E. Lee, T. Park, Shape-stabilized orange peel/myristic acid phase change materials for efficient thermal energy storage application, Energy Rep. 8 (2022) 9618–9628, https://doi.org/10.1016/j.egyr.2022.07.143.

[5] C. Li, Q. Li, R. Ge, A review of heat transfer performance enhancement and applications of inorganic salt based shape-stabilized composite phase change materials for medium and high temperature thermal energy storage, Energy Rep. 8 (2022) 12740–12764, https://doi.org/10.1016/j.egyr.2022.09.073.

[6] T. Yamamoto, S. Hatayama, Y. Sutou, Design strategy of phase change material properties for low-energy memory application, Mater. Des. 216 (2022) 110560, https://doi.org/10.1016/j.matdes.2022.110560.

[7] K. Faraj, M. Khaled, J. Faraj, F. Hachem, K. Chahine, C. Castelain, Short recent summary review on evolving phase change material encapsulation techniques for building applications, Energy Rep. 8 (2022) 1245–1260, https://doi.org/10.1016/j.egyr.2022.07.135.

[8] Z. Liu, M. Xie, Y. Zhou, Y. He, L. Zhang, G. Zhang, D. Chen, A state-of-the-art review on shallow geothermal ventilation systems with thermal performance enhancement: system classifications, advanced technologies and applications, Energy Built Environ. 4 (2021) 148–168, https://doi.org/10.1016/j.enbenv.2021.10.003.

[9] A.K. Azad, S. Parvin, Bibliometric analysis of photovoltaic thermal (PV/T) system: from citation mapping to research agenda, Energy Rep. 8 (2022) 2699–2711, https://doi.org/10.1016/j.egyr.2022.01.182.

[10] A.M. Nair, C. Wilson, M.J. Huang, P. Griffiths, N. Hewitt, Phase change materials in building integrated space heating and domestic hot water applications: a review, J. Energy Storage 54 (2022) 105227, https://doi.org/10.1016/j.est.2022.105227.

[11] A. Chibani, S. Merouani, F. Benmoussa, Computational analysis of the melting process of phase change material-metal foam-based latent thermal energy storage unit: the heat exchanger configuration, J. Energy Storage 42 (2021) 103071, https://doi.org/10.1016/j.est.2021.103071.

[12] S. Rostami, M. Afrand, A. Shahsavar, M. Sheikholeslami, R. Kalbasi, S. Aghakhani, M.S. Shadloo, H.F. Oztop, A review of melting and freezing processes of PCM/nano-PCM and their application in energy storage, Energy. 211 (2020) 118698, https://doi.org/10.1016/j.energy.2020.118698.

[13] S. Zhang, D. Feng, L. Shi, L. Wang, Y. Jin, L. Tian, Z. Li, G. Wang, L. Zhao, Y. Yan, A review of phase change heat transfer in shape-stabilized phase change materials (ss-PCMs) based on porous supports for thermal energy storage, Renew. Sust. Energ. Rev. 135 (2021) 110127, https://doi.org/10.1016/j.rser.2020.110127.

[14] F.L. Rashid, A. Rahbari, R.K. Ibrahem, P. Talebizadehsardari, A. Basem, A. Kaood, H.I. Mohammed, M.H. Abbas, M.A. Al-Obaidi, Review of solidification and melting performance of phase change materials in the presence of magnetic field, rotation, tilt angle, and vibration, J. Energy Storage 67 (2023) 107501, https://doi.org/10.1016/j.est.2023.107501.

[15] M. Tao, L. Zhenpeng, Z. Jiaxin, Photovoltaic panel integrated with phase change materials (PV-PCM): technology overview and materials selection, Renew. Sust. Energ. Rev. 116 (2019) 109406, https://doi.org/10.1016/j.rser.2019.109406.

[16] M.M. Farid, A.M. Khudhair, S.A.K. Razack, S. Al-Hallaj, A review on phase change energy storage: materials and applications, Energy Convers. Manag. 45 (2004) 1597–1615, https://doi.org/10.1016/j.enconman.2003.09.015.

[17] A. Prieto, U. Knaack, T. Auer, T. Klein, COOLFACADE: state-of-the-art review and evaluation of solar cooling technologies on their potential for façade integration, Renew. Sust. Energ. Rev. 101 (2019) 395–414, https://doi.org/10.1016/j.rser.2018.11.015.

[18] P. Kohlhepp, H. Harb, H. Wolisz, S. Waczowicz, D. Müller, V. Hagenmeyer, Large-scale grid integration of residential thermal energy storages as demand-side flexibility resource: a review of international field studies, Renew. Sust. Energ. Rev. 101 (2019) 527–547, https://doi.org/10.1016/j.rser.2018.09.045.

[19] A. Miglioli, N. Aste, C. Del Pero, F. Leonforte, Photovoltaic-thermal solar-assisted heat pump systems for building applications: integration and design methods, Energy Built Environ. 4 (2022) 39–56, https://doi.org/10.1016/j.enbenv.2021.07.002.

[20] M.K. Sharma, Alternative designs and technological advancements of phase change material integrated photovoltaics: a state-of-the-art review, J. Energy Storage 48 (2022) 104020, https://doi.org/10.1016/j.est.2022.104020.

[21] Z. Xu, Q. Kong, H. Qu, C. Wang, Case studies in thermal engineering cooling characteristics of solar photovoltaic panels based on phase change materials, Case Stud. Therm. Eng. 41 (2023) 102667, https://doi.org/10.1016/j.csite.2022.102667.

[22] M. Lotfi, A.H. Shiravi, M. Firoozzadeh, Experimental study on simultaneous use of phase change material and reflector to enhance the performance of photovoltaic modules, J. Energy Storage 54 (2022) 105342, https://doi.org/10.1016/j.est.2022.105342.

[23] M. Jurčević, S. Nižetić, D. Čoko, M. Arıcı, A.T. Hoang, E. Giama, A. Papadopoulos, Techno-economic and environmental evaluation of photovoltaic-thermal collector design with pork fat as phase change material, Energy 254 (2022), https://doi.org/10.1016/j.energy.2022.124284.

[24] M.A. Al-Nimr, A.I. Dawahdeh, J.A. Al-Omari, Dual power generation modes for thermally regenerative electrochemical cycle integrated with concentrated thermal photovoltaic and phase change material storage, J. Energy Storage 58 (2023) 106373, https://doi.org/10.1016/j.est.2022.106373.

[25] A. Yusuf, S. Ballikaya, Performance analysis of concentrated photovoltaic systems using thermoelectric module with phase change material, J. Energy Storage 59 (2023) 106544, https://doi.org/10.1016/j.est.2022.106544.

[26] C. Zhang, N. Wang, H. Xu, Y. Fang, Q. Yang, F.K. Talkhoncheh, Thermal management optimization of the photovoltaic cell by the phase change material combined with metal fins, Energy. 263 (2023) 125669, https://doi.org/10.1016/j.energy.2022.125669.

[27] Z. Deng, X. Liu, C. Zhang, Y. Huang, Y. Chen, Melting behaviors of PCM in porous metal foam characterized by fractal geometry, Int. J. Heat Mass Transf. 113 (2017) 1031–1042, https://doi.org/10.1016/j.ijheatmasstransfer.2017.05.126.

[28] A. Mourad, N.A.A. Qasem, A. Abderrahmane, R. Marzouki, K. Guedri, O. Younis, N. A. Shah, T. Botmart, Numerical study on n-octadecane PCM melting process inside a pear-shaped finned container, Case Stud. Therm. Eng. 38 (2022) 102328, https://doi.org/10.1016/j.csite.2022.102328.

[29] Y.M.F.E. Hasadi, J.M. Khodadadi, Numerical simulation of solidification of colloids inside a differentially heated cavity, J. Heat Transf. 137 (2015) 1–10, https://doi.org/10.1115/1.4029035.
[30] H. Senobar, M. Aramesh, B. Shabani, Nanoparticles and metal foams for heat transfer enhancement of phase change materials: a comparative experimental study, J. Energy Storage 32 (2020) 101911, https://doi.org/10.1016/j.est.2020.101911.
[31] C.J. Ho, J.Y. Gao, Preparation and thermophysical properties of nanoparticle-in-paraffin emulsion as phase change material, Int. Commun. Heat Mass Transf. 36 (2009) 467–470, https://doi.org/10.1016/j.icheatmasstransfer.2009.01.015.
[32] A. Chibani, S. Merouani, H. Laidoudi, A. Dehane, M.R. Morakchi, L. Bendada, Analysis and optimization of concentrator photovoltaic system using a phase change material (RT 35HC) combined with variable metal fins, J. Energy Storage 72 (2023) 108283, https://doi.org/10.1016/j.est.2023.108283.
[33] A. Chibani, S. Merouani, Acceleration of heat transfer and melting rate of a phase change material by nanoparticles addition at low concentrations, Int. J. Thermophys. 42 (2021) 1–16, https://doi.org/10.1007/s10765-021-02822-z.
[34] A. Chibani, S. Merouani, F. Benmoussa, M.H. Abdellattif, A. Erto, B.H. Jeon, Y. Benguerba, A strategy for enhancing heat transfer in phase change material-based latent thermal energy storage unit via nano-oxides addition: a study applied to a shell-and-tube heat exchanger, J. Environ. Chem. Eng. 9 (2021) 106744, https://doi.org/10.1016/j.jece.2021.106744.
[35] T. Alqahtani, S. Mellouli, A. Bamasag, F. Askri, P.E. Phelan, Thermal performance analysis of a metal hydride reactor encircled by a phase change material sandwich bed, Int. J. Hydrog. Energy 45 (2020) 23076–23092, https://doi.org/10.1016/j.ijhydene.2020.06.126.
[36] J.M. Mahdi, E.C. Nsofor, Solidification enhancement of PCM in a triplex-tube thermal energy storage system with nanoparticles and fins, Appl. Energy 211 (2018) 975–986, https://doi.org/10.1016/j.apenergy.2017.11.082.
[37] J.M. Mahdi, H.I. Mohammed, P. Talebizadehsardari, M. Ghalambaz, H.Sh. Majdi, A. Khan, W. Yaïci, D. Giddings, Simultaneous and consecutive charging and discharging of a PCM-based domestic air heater with metal foam, Appl. Therm. Eng. 197 (2021), 117408, https://doi.org/10.1016/j.applthermaleng.2021.117408.
[38] A.A. Al-Abidi, S. Mat, K. Sopian, M.Y. Sulaiman, A.T. Mohammad, Numerical study of PCM solidification in a triplex tube heat exchanger with internal and external fins, Int. J. Heat Mass Transf. 61 (2013) 684–695, https://doi.org/10.1016/j.ijheatmasstransfer.2013.02.030.
[39] J.M. Mahdi, E.C. Nsofor, Solidification of a PCM with nanoparticles in triplex-tube thermal energy storage system, Appl. Therm. Eng. 108 (2016) 596–604, https://doi.org/10.1016/j.applthermaleng.2016.07.130.
[40] Z. Khan, Z. Ahmad Khan, Experimental and numerical investigations of nano-additives enhanced paraffin in a shell-and-tube heat exchanger: a comparative study, Appl. Therm. Eng. 143 (2018) 777–790, https://doi.org/10.1016/j.applthermaleng.2018.07.141.
[41] Z. Khan, Z.A. Khan, P. Sewell, Heat transfer evaluation of metal oxides based nano-PCMs for latent heat storage system application, Int. J. Heat Mass Transf. 144 (2019) 118619, https://doi.org/10.1016/j.ijheatmasstransfer.2019.118619.
[42] M.A. Aichouni, N.F. Alshammari, N. Ben Khedher, M. Aichouni, Experimental Evaluation of Nano-enhanced Phase Change Materials in a Finned Storage Unit 10, 2020, pp. 5814–5818.
[43] A. Kouravand, A. Kasaeian, F. Pourfayaz, M.A. Vaziri Rad, Evaluation of a nanofluid-based concentrating photovoltaic thermal system integrated with finned PCM heatsink: an experimental study, Renew. Energy 201 (2022) 1010–1025, https://doi.org/10.1016/j.renene.2022.11.025.
[44] M. Gorzin, M.J. Hosseini, M. Rahimi, R. Bahrampoury, Nano-enhancement of phase change material in a shell and multi-PCM-tube heat exchanger, J. Energy Storage 22 (2019) 88–97, https://doi.org/10.1016/j.est.2018.12.023.
[45] I. Sarani, S. Payan, S.A. Nada, A. Payan, Numerical investigation of an innovative discontinuous distribution of fins for solidification rate enhancement in PCM with and without nanoparticles, Appl. Therm. Eng. 176 (2020) 115017, https://doi.org/10.1016/j.applthermaleng.2020.115017.
[46] C. Ji, Z. Qin, Z. Low, S. Dubey, F.H. Choo, F. Duan, Non-uniform heat transfer suppression to enhance PCM melting by angled fins, Appl. Therm. Eng. 129 (2018) 269–279, https://doi.org/10.1016/j.applthermaleng.2017.10.030.
[47] F. Zhu, C. Zhang, X. Gong, Numerical analysis and comparison of the thermal performance enhancement methods for metal foam/phase change material composite, Appl. Therm. Eng. 109 (2016) 373–383, https://doi.org/10.1016/j.applthermaleng.2016.08.088.
[48] Z. Liu, Y. Yao, H. Wu, Numerical modeling for solid-liquid phase change phenomena in porous media: Shell-and-tube type latent heat thermal energy storage, Appl. Energy 112 (2013) 1222–1232, https://doi.org/10.1016/j.apenergy.2013.02.022.
[49] A.A. Rabienataj Darzi, M. Jourabian, M. Farhadi, Melting and solidification of PCM enhanced by radial conductive fins and nanoparticles in cylindrical annulus, Energy Convers. Manag. 118 (2016) 253–263, https://doi.org/10.1016/j.enconman.2016.04.016.
[50] M.S. Mahdi, H.B. Mahood, A.N. Campbell, A.A. Khadom, Natural convection improvement of PCM melting in partition latent heat energy storage: numerical study with experimental validation, Int. Commun. Heat Mass Transf. 126 (2021) 105463, https://doi.org/10.1016/j.icheatmasstransfer.2021.105463.
[51] F.L. Tan, S.F. Hosseinizadeh, J.M. Khodadadi, L. Fan, Experimental and computational study of constrained melting of phase change materials (PCM) inside a spherical capsule, Int. J. Heat Mass Transf. 52 (2009) 3464–3472, https://doi.org/10.1016/j.ijheatmasstransfer.2009.02.043.
[52] R. Parthiban, P. Ponnambalam, An enhancement of the solar panel efficiency: a comprehensive review, Front. Energy Res. 10 (2022) 1–15, https://doi.org/10.3389/fenrg.2022.937155.
[53] C.Y. Zhao, W. Lu, Y. Tian, Heat transfer enhancement for thermal energy storage using metal foams embedded within phase change materials (PCMs), Sol. Energy 84 (2010) 1402–1412, https://doi.org/10.1016/j.solener.2010.04.022.
[54] A.V. Arasu, A.S. Mujumdar, Numerical study on melting of paraffin wax with Al2O3 in a square enclosure, Int. Commun. Heat Mass Transf. 39 (2012) 8–16, https://doi.org/10.1016/j.icheatmasstransfer.2011.09.013.
[55] A.R. Abdulmunem, P.M. Samin, H.A. Rahman, H.A. Hussien, I.I. Mazali, H. Ghazali, Experimental and numerical investigations on the effects of different tilt angles on the phase change material melting process in a rectangular container, J. Energy Storage 32 (2020) 101914, https://doi.org/10.1016/j.est.2020.101914.
[56] S. Lv, J. Yang, J. Ren, B. Zhang, Y. Lai, Z. Chang, Research and numerical analysis on performance optimization of photovoltaic-thermoelectric system incorporated with phase change materials, Energy. 263 (2023) 125850, https://doi.org/10.1016/j.energy.2022.125850.
[57] M. Emam, S. Ookawara, M. Ahmed, Thermal management of electronic devices and concentrator photovoltaic systems using phase change material heat sinks: experimental investigations, Renew. Energy 141 (2019) 322–339, https://doi.org/10.1016/j.renene.2019.03.151.
[58] M.R. Hajizadeh, A.N. Keshteli, Q.V. Bach, Solidification of PCM within a tank with longitudinal-Y shape fins and CuO nanoparticle, J. Mol. Liq. 317 (2020) 114188, https://doi.org/10.1016/j.molliq.2020.114188.